

Class recording (before mid): covering Graph representation, BFS, DFS, Shortest path, Edge classification. [Class recording link](#)

****ALL CLASS RECORDING PLAYLIST FOR ONLINE SESSIONS OF THIS SEMESTER:**
[ONLINE CLASSES HFN**](#)

Edge classification supplementary: [DFS - Types of Edges | Edge Classification | Tree Edge, Back Edge, Forward Edge, Cross Edge](#)

Traversal comparison: [A Comparison of Pathfinding Algorithms](#) [SKIP A* Algorithm for now]

Slide: [Graph\[BUX\].pdf](#)

How to check bipartite graph: [Bipartite Graph \(BFS\) | Graph Coloring](#)

How to find Cycle in graph: (DFS method) [Detect cycle in a directed graph](#)

Topological sort: (DFS) [Topological Sort Algorithm | Graph Theory](#)
(BFS/ Kahn's ALgo) [Topological Sort | Kahn's Algorithm | Graph Theory](#)

Strongly Connected Components:

- 1 [Kosaraju Algorithm | Strongly connected components in a graph](#)
- 2 [Tarjans strongly connected components algorithm](#)

Single source shortest path algorithms:

1. [Dijkstra's Shortest Path Algorithm | Graph Theory](#)
2. [Bellman Ford Algorithm | Shortest path & Negative cycles | Graph Theory](#)
3. [Shortest/Longest path on a Directed Acyclic Graph \(DAG\) | Graph Theory](#)

EXTRA CLASS: **Tuesday 5th Dec [7.30 to 9.30]**

Assignments: 3 [S1, S2, S3] **s1: BFS,DFS,S1**

Quiz: S1, S2/S3

Syllabus:

1. Single source shortest path Algo: **Dijkstra**, Bellman-Ford [1]
2. Greedy Algo: MST(Minimum Spanning Tree: Prims/**Kruskal(DSU)**) [2]
3. Huffman Encoding.
4. Dynamic Programming: Knapsack + LCS [2]

5. P vs NP [1]